MWALIMU EXAMINATIONS BUREAU

End of Term 1 Examinations-2017

CHEMISTRY P1 MARKING GUIDE

1.(a) Is the enthalpy change (heat evolved or absorbed) which occurs when one mole of a compound (in its standard state) is formed from its constituent elements in their standard physical states under standard conditions 01

(b) I. $C(s) + 2H_2(g) \longrightarrow CH_4(g)$ $\Delta H_1^\theta = -76\ kJ\ mol^{-1}$

II. $C(s) + O_2(g) \longrightarrow CO_2(g)$ $\Delta H_2^\theta = -394\ kJ\ mol^{-1}$ 04

III. $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$ $\Delta H_3^\theta = -286\ kJ\ mol^{-1}$

IV $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$ $\Delta H_c^\theta = ?$

ΔH_c^θ = (Reversed I) + (II) + 2(III)

$= (+76) + (-394) + (-286 \times 2) = -890\ kJ\ mol^{-1}$

accept $\Delta H_c^\theta = \sum \Delta H_f^\theta (\text{products}) - \sum \Delta H_f^\theta (\text{reactants})$

$= (\Delta H_f^\theta (CO_2) + 2\Delta H_f^\theta (H_2O)) - (\Delta H_f^\theta (CH_4) + 2\Delta H_f^\theta (O_2))$

$= (-394) + 2(-286) - (-76) - 2(0)$

$= -890\ kJ\ mol^{-1}$

2 (a)(i) +7 $0\frac{1}{2}$

(ii) +6 $0\frac{1}{2}$

(iii) +4 $0\frac{1}{2}$

(b)(i) $MnO_2(s) + 4HCl(aq) \longrightarrow MnCl_2(aq) + Cl_2(g) + 2H_2O(l)$

Black solid/powder dissolves with effervescence 02

any two giving greenish-yellow gas and a colourless solution

(ii) $3MnO_4^{2-}(aq) + 2H_2O(l) \longrightarrow 2MnO_4^-(aq) + MnO_2(s) + 4OH^-(aq)$

The green solution turns purple and a black solid is formed

3 (a) $\longrightarrow CH_2{=}CH_2 + H_2O$

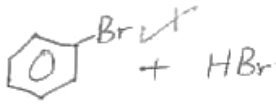
mechanism

let alcohol be ROH

$ROH + KOH \longrightarrow RO^- + K^+ + H_2O(l)$

$CH_2-CH_2Br \longrightarrow CH_2=CH_2 + Br^- + ROH$ 02

(for all correct)

(b) 

mechanism; $2Fe_{(s)} + 3Br_2(l) \longrightarrow 2FeBr_3(s)$ balanced

$Br-Br \; FeBr_3 \rightleftharpoons Br^+ + FeBr_4^-$

$(H^+ + FeBr_4^- \longrightarrow HBr + FeBr_3)$ 3½

4(a)(i) $^{233}_{91}Pa \longrightarrow \beta + ^{233}_{92}U \longrightarrow ^{4}_{2}He + ^{229}_{90}Th$ 02

(ii) $^{239}_{94}Pu \longrightarrow ^{91}_{39}Y + ^{146}_{55}Cs + 2\,^{1}_{0}n$ 01

(b) $t_{\frac{1}{2}} = \frac{\ln 2}{k} \Rightarrow k = \left(\frac{\ln 2}{20}\right)$ minute^{-1}

$\log\left(\frac{N_0}{N_t}\right) = kt \Rightarrow \log\left(\frac{N_0}{N_t}\right) = \left(\frac{\ln 2}{20}\right)(70) \Rightarrow \frac{N_0}{N_t} = 10^{2.426015132}$ 2½

$\frac{N_t}{N_0} = 3.749599374 \times 10^{-3}$

Required percentage $= \frac{N_t}{N_0} \times 100 = 0.37\%$

5(a)(i) $Cl_2(g) + 2Br^-(aq) \longrightarrow 2Cl^-(aq) + Br_2(aq)$ ignore state 01

or $Cl_2(g) + 2KBr(aq) \longrightarrow 2KCl(aq) + Br_2(aq)$

(ii) $Cl_2(g) + 2OH^-(aq) \longrightarrow ClO^-(aq) + Cl^-(aq) + H_2O(l)$ ignore state 01

or $Cl_2(g) + 2NaOH(aq) \longrightarrow NaClO(aq) + NaCl(aq) + H_2O(l)$

(b)(i) $E^{\theta}_{cell} = E^{\theta}_{right} - E^{\theta}_{left} = (+1.36) - (-0.771) = +2.131V$ 2½

Thus the reaction is feasible since E^{θ}_{cell} is positive

(ii) The green solution turns to yellow/brown 1½

6 (a) Reagent: Aqueous iodine solution and aqueous sodium hydroxide solution 02

observations: CH_3OH gives no observable change

CH_3CH_2OH gives a(pale) yellow precipitate/solid

(b) Reagent: Ammoniacal silver nitrate solution

observations: $CH_3C\equiv CCH_3$ gives no observable change 02

$HC\equiv CCH_2CH_3$ gives a white precipitate

<u>allow</u> Ammoniacal copper(I) chloride solution

$CH_3C\equiv CCH_3$ gives no observable change

$HC\equiv CCH_2CH_3$ gives a reddish-brown/brown precipitate/solid

7. (a) If a solute is shaken with two immiscible solvents in the same container, the solute will distribute itself between the two solvents such that the concentration of the solute in one solvent divided by the concentration of the solute in the second solvent is always a constant at equilibrium provided the solute remains in the same molecular state and temperature is kept constant. 02

<u>or</u>

$$K_D = \frac{\text{concentration of a solute X in solvent A}}{\text{concentration of a solute X in solvent B}}$$

Solvents A and B are immiscible in the same container at equilibrium at constant temperature and solute X remains in the same molecular state in both solvents A and B.

(b) let x moles be in the aqueous layer

$\Rightarrow$ (0.5−x) moles go into benzene 02

$\therefore \left(\frac{x}{60}\right) \Big/ \left(\frac{0.5-x}{30}\right) = 0.4 \Rightarrow x = 0.4 - 0.8x$

$1.8x = 0.4$

$x = \frac{0.4}{1.8} = 0.22$

8(a)(i) Magnesium reacts slowly with cold water forming magnesium hydroxide and hydrogen 03

$Mg(s) + H_2O(l) \longrightarrow Mg(OH)_2(aq) + H_2(g)$

Red hot clean magnesium burns in steam readily to form magnesium oxide and hydrogen

$Mg(s) + H_2O(g) \longrightarrow MgO(s) + H_2(g)$

(ii) Barium reacts vigorously with cold water to form barium hydroxide and hydrogen 1½

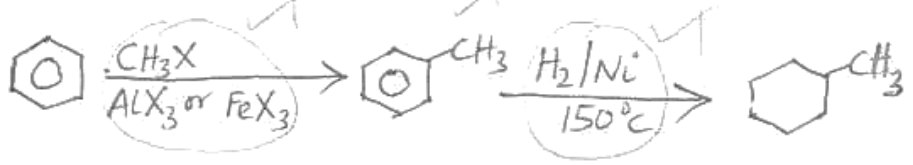
$Ba(s) + H_2O(l) \longrightarrow Ba(OH)_2(aq) + H_2(g)$

It also burns in steam forming barium oxide and hydrogen.

$Ba(s) + H_2O(g) \longrightarrow BaO(s) + H_2(g)$ (none)

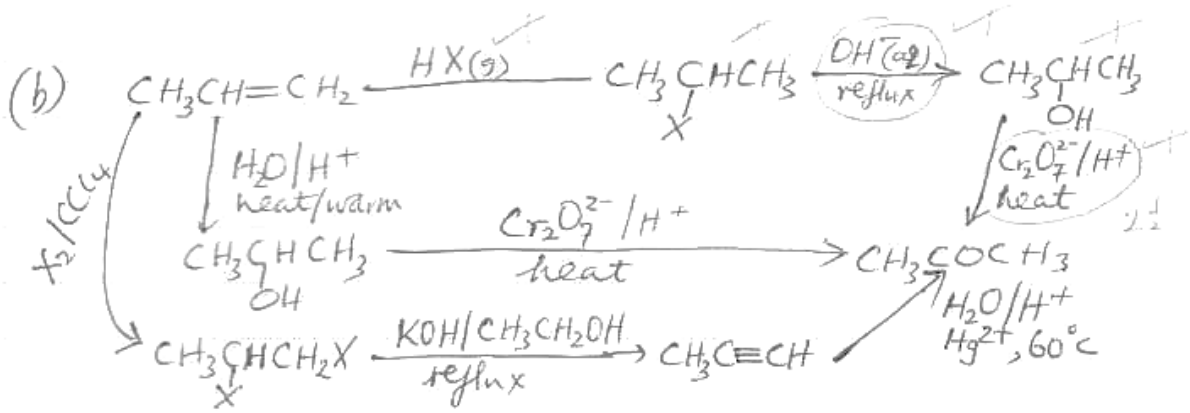
(b)(i) $Be_2C(s) + 4H_2O(l) \longrightarrow 2Be(OH)_2(s) + CH_4(g)$ 01

(ii) $CaC_2(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(aq) + C_2H_2(g)$ 01

9(a) 

1½

(b)



10 (a)(i) % of H in H_2O formed $= \left(\frac{2}{18} \times \frac{4.5}{3.7} \times 100\right)\% = 13.51\%$

volume of CO_2 formed $= (6.48 - 2.0)\,dm^3 = 4.46\,cm^3$

mass of CO_2 formed $= \left(\frac{4.46}{22.4} \times 44\right) g = 8.76\,g$

% of C in CO_2 formed $= \left(\frac{12}{44} \times \frac{8.76}{3.7} \times 100\right)\% = 64.57\%$

% of O in Q $= 100 - (64.57 + 13.51) = 21.92\%$

Elements	C :	H	O
No of moles	$\frac{64.57}{12}$	$\frac{13.51}{1}$	$\frac{21.92}{16}$
	5.38	13.51	1.37
atomic ratio	5.38/1.37	13.51/1.37	1.37/1.37
	4	10	1

03

EF of Q is $C_4H_{10}O$

(ii) let $(C_4H_{10}O)_n = 74 \Rightarrow 74n = 74;\ n = 1$

MF of Q is $C_4H_{10}O$

01

(b) $CH_3CH_2CH_2CH_2OH$ $CH_3CH_2OCH_2CH_3$

$CH_3CH_2CHCH_3$ (OH) $CH_3CH_2CH_2OCH_3$

$(CH_3)_3C-OH$ $(CH_3)_2CHCH_2OH$

02

any 4 correct

(c) P; CH_3CH_2COONa (or $CH_3CH_2COO^-$) R; CH_3CH_2COOH

Z; CH_3CH_2COCl

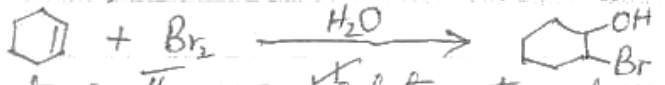
03

Isomer must be cited in (b)

11(a) observation: The orange solution turned green 2½

Equation: $Cr_2O_7^{2-} + 6I^-_{(aq)} + 14H^+_{(aq)} \rightarrow 2Cr^{3+}_{(aq)} + 3I_2(aq) + 7H_2O(l)$

(b) observation; The reddish-brown solution turned colourless

Equation: 

02

(c) observation: The green solution turned yellow 2½

Equation; $H_2O_2(aq) + 2H^+_{(aq)} + 2Fe^{2+}_{(aq)} \rightarrow 2H_2O(l) + 2Fe^{3+}_{(aq)}$

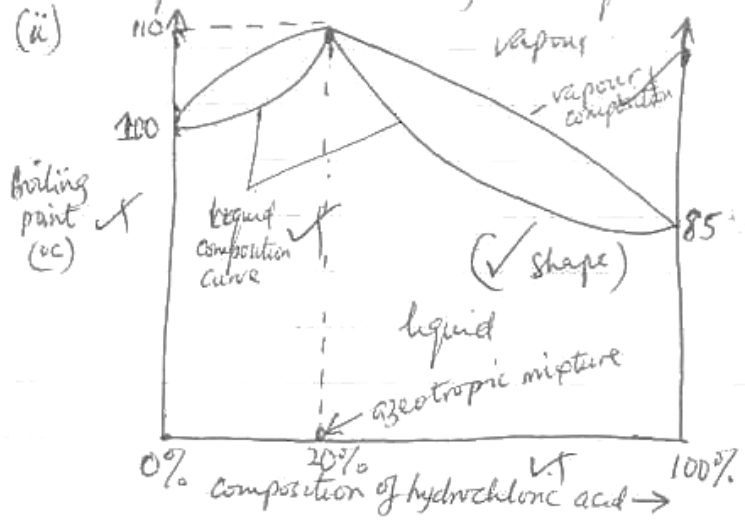
(d) observation: The brown turnings/solid dissolved with reddish brown/brown fumes (gas) and a green/blue solution is formed. any one correct observation 02

Equation; $Cu(s) + 4HNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2H_2O(l) + 2NO_2(g)$

12(a) Is a solution in which the intermolecular forces between the unlike molecules (A–B) and the like molecules (A–A) and (B–B) are equal/similar. The solution is formed without change in heat/temperature and volume. The solution obeys Raoult's law (of vapour pressure). 02

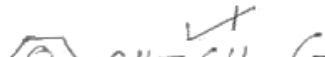
(b)(i) A mixture of two liquids that boils at constant temperature at a given pressure to give a vapour of the same composition as that of the liquid. 01

(ii)

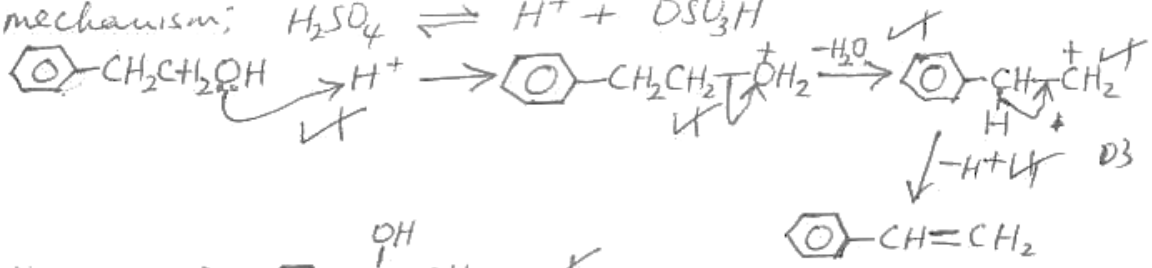


03

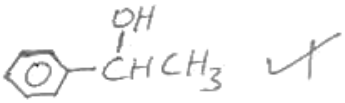
(iii) When a mixture containing 10% hydrochloric acid is fractionally distilled, it boils to give a vapour which is richer in water than the original liquid mixture. Cooling this vapour $2\frac{1}{2}$ produces a liquid mixture of the same composition which when heated produces a vapour richer in water than itself. Repeated heating and cooling finally produces pure water as distillate and the azeotropic mixture as the residue

13 (a) ⟶  (+ H_2O) (ignore H_2O)

mechanism; $H_2SO_4 \rightleftharpoons H^+ + \bar{O}SO_3H$



03

(b) ⟶ 

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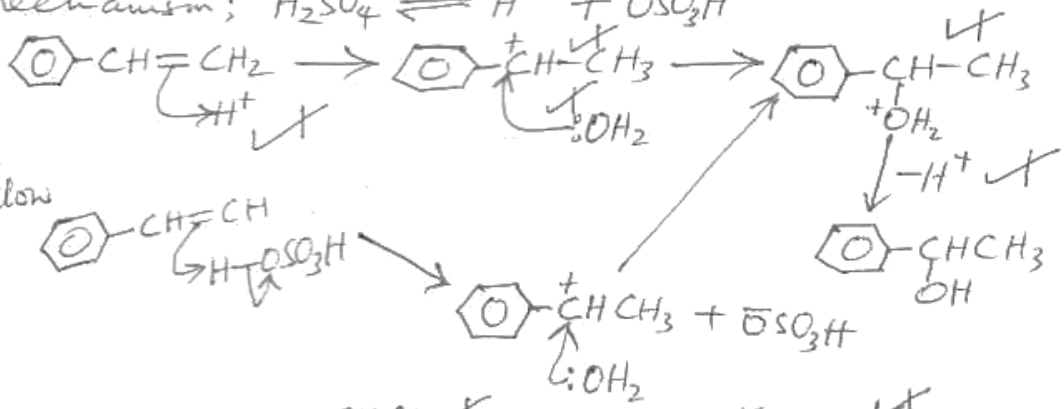
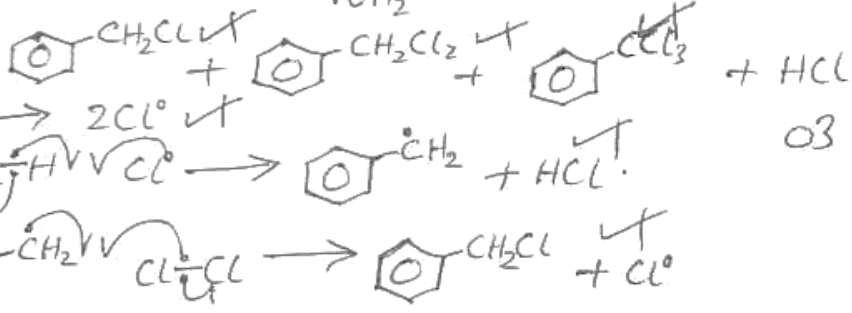


03

(c) ⟶ 

03

14(a)(i) – Both form monoxides when burnt in oxygen ✓
– Both the hydroxides of lithium and magnesium decompose when heated to give the corresponding oxide and water ✓
– Their nitrates decompose on heating to give the corresponding oxide in addition to nitrogen dioxide and oxygen ✓
– Their carbonates and phosphates are sparingly soluble in water
– Their hydrogencarbonates exist only in solution. ✓ 04
– Both form ionic nitrides when burnt in molecular nitrogen.
– Their halides (except fluorides) are soluble in organic solvents.

any correct 4 – 01 for extra wrong – 0½ for extra correct

(ii) • Their atoms have the same electronegativity ✓ 01
• Their atoms have the same electrode potential
• Their cations have the same charge density/polarising power

(iii) • silicon and boron ✓ rej wrong spelling
• aluminium and beryllium

(b)(i) $Mg(s) + 2HCl(g) \longrightarrow MgCl_2(s) + H_2(g)$ ✓✗ 1½
or $Mg(s) + Cl_2(g) \longrightarrow MgCl_2(s)$

(ii) $MgCl_2 \cdot 6H_2O(l) \longrightarrow Mg(OH)Cl(s) + HCl(g) + 5H_2O(l)$ ✓✗ 1½

15 (a)(i) $C_6H_5COO^-(aq) + H_2O(l) \rightleftharpoons C_6H_5COOH(aq) + OH^-(aq)$ ✓ 01

or $C_6H_5COO^-Na^+(aq) + H_2O(l) \rightleftharpoons C_6H_5COOH(aq) + NaOH(aq)$ $-0\frac{1}{2}$ for ⇌ / states

accept ⬡ for C_6H_5-

(ii) $K_h = \frac{[C_6H_5COOH][OH^-]}{[C_6H_5COO^-]}$ ✓ 01

(b) Rfm of $C_6H_5COO^- = (12\times7)+(1\times5)+(2\times16) = 121$ ✓

$[C_6H_5COO^-] = \frac{24.2}{121} = 0.2\ mol\ dm^{-3}$ ✓

At equilibrium; $[OH^-] = [C_6H_5COO^-]$ ✓

$K_h = \frac{[OH^-]^2}{[C_6H_5COO^-]}$ ✓ ; $1.6\times10^{-10} = \frac{[OH^-]^2}{0.2}$ ✓ $4\frac{1}{2}$

$[OH^-] = \sqrt{1.6\times10^{-10}\times0.2}$ ✓

$= 5.657\times10^{-6}\ mol\ dm^{-3}$ ✓

$K_w = [OH^-][H^+]$; $[H^+] = \frac{1.0\times10^{-14}}{5.657\times10^{-6}}$ ✓ $= 1.768\times10^{-9}\ M$

$pH = -\log[H^+]$ ✓ $= -\log(1.768\times10^{-9})$ ✓

$= 8.75$ units ✓

(c) The pH will remain constant ✓

This is because the solution contains a large concentration of un-ionised benzoic acid molecules ✓. Addition of sodium hydroxide to this solution causes a reaction between hydroxyl ions from sodium hydroxide and the benzoic acid molecules which keeps the pH constant. ✓ $2\frac{1}{2}$

$OH^-(aq) + C_6H_5COOH(aq) \longrightarrow C_6H_5COO^-(aq) + H_2O(l)$ ✓

16 (a) Ammonia is a weak base which ionises partially in water/aqueous solution to form ammonium ions and hydroxide ions. ✓

$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$ ✓

When added to a solution containing aluminium ions, the product of the concentration of aluminium ions and concentration of hydroxide ions exceeds the

solubility product of aluminium hydroxide hence precipitating aluminium hydroxide.

$Al^{3+}(aq) + 3OH^-(aq) \rightarrow Al(OH)_3(s)$ ✓ 05

In the presence of ammonium ions, the ionisation of ammonia is suppressed due to common ion effect. Thus the concentration of hydroxyl ions reduces to very low values that the solubility product of aluminium hydroxide is not exceeded by its ionic product. Therefore no precipitate is seen.

(b) For group II elements, each atom contributes only two electrons towards metallic bond formation whereas for group I elements, each atom contributes one electron. As a result, group II elements form stronger metallic bonds than the corresponding group I elements. The bonds in group II elements therefore require more energy to break hence higher 04 melting points compared to those of the corresponding group I elements. Also elements of group II have much smaller atomic/metallic radii than those of the corresponding group I elements. Thus their atoms closely pack themselves than the atoms of group I elements.

17 (a) – Roasting zinc sulphide / iron disulphide in air
– burning sulphur in oxygen 01

(b) (i) The position of equilibrium shifts from left to the right. This is because the forward reaction proceeds with decrease in volume. Thus oxygen $1\frac{1}{2}$ and sulphur dioxide react to form sulphur trioxide as a result of increase in pressure so as to decrease volume

(ii) The position of the equilibrium shifts from right to the left. This is because the forward reaction is exothermic favoured by reduction in temperature. $1\frac{1}{2}$

(iii) The equilibrium position remains unchanged. This is because vanadium (V) oxide is a catalyst and affects both forward and backward reactions equally. $1\frac{1}{2}$

(c)(i) moles of O_2 that reacted $= 0.5$

moles of O_2 that remained $= 0.8 - 0.5 = 0.3$

$[O_2] = \frac{0.3}{2} = 0.15\ mol\ l^{-1}$

moles of SO_2 that reacted $= 1$

moles of SO_2 that remained $= 1.4 - 1 = 0.4$

$[SO_2] = \frac{0.4}{2} = 0.2\ mol\ l^{-1}$

(ii) $K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]} = \frac{(\frac{1}{2})^2}{(0.2)^2(0.15)} = 4.17\ mol^{-1}l$ $1\frac{1}{2}$

End